Estimation with informative missing data in the low-rank case

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Classical definitions

- ⋆ $Y \in \mathbb{R}^{n \times p}$ the data matrix,
- ⋆ $M \in \mathbb{R}^{n \times p}$ the missing-data pattern:

$$M_{ij} = \begin{cases} 1 & \text{if } Y_{ij} \text{ is observed,} \\ 0 & \text{otherwise,} \end{cases}$$

- ⋆ Y_{obs} the observed variables, Y_{mis} the missing variables.
- ⋆ **Issue: Cause of the missingness ?**
  [Rubin, 1976], [Little and Rubin, 2014]

Toy Example:

$$\begin{array}{cc} Y_1 & Y_2 \\ \end{array}$$
$$\begin{pmatrix} 1 & 2 \\ 3 & 20 \\ 22 & 4 \end{pmatrix}$$

MCAR mechanism

$$p(M|Y;\phi) = p(M;\phi), \quad \forall Y, \phi.$$

ϕ: the unknown parameters of the missingness.

MAR mechanism

$$p(M|Y;\phi) = p(M|Y_{\text{obs}};\phi), \quad \forall Y_{\text{mis}}, \phi.$$

MNAR mechanism

Other case, i.e.

$$p(M|Y;\phi) = p(M|Y_{\text{obs}}, Y_{\text{mis}};\phi), \quad \forall \phi.$$

MCAR

$$\begin{array}{cc} Y_1 & Y_2 \\ \end{array}$$
$$\begin{pmatrix} \text{NA} & 2 \\ 3 & 20 \\ 22 & 4 \end{pmatrix}$$

MAR

$$\begin{array}{cc} Y_1 & Y_2 \\ \end{array}$$
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MNAR

$$\begin{array}{cc} Y_1 & Y_2 \\ \end{array}$$
$$\begin{pmatrix} 1 & 2 \\ 3 & 20 \\ \text{NA} & 4 \end{pmatrix}$$

$\rightsquigarrow$ self-masked MNAR.

Likelihood approach

- Θ: unknown parameter.
- Likelihood-approach: maximizing the joint log-likelihood,

$$\ell(\Theta, \phi; Y, M) = p(Y; \Theta) p(M|Y; \phi).$$

- Missing data: maximizing the observed joint log-likelihood,

$$\ell(\Theta, \phi; Y_{\text{obs}}, M) = \int \ell(\Theta, \phi; Y, M) dY_{\text{mis}}.$$

✓ In the MAR setting, one can ignore the mechanism since

$$p(M|Y; \phi) = p(M|Y_{\text{obs}}; \phi).$$

$$\Rightarrow \ell(\Theta, \phi; Y_{\text{obs}}, M) \propto \ell(\Theta; Y_{\text{obs}}) = \int \ell(\Theta; Y) dY_{\text{mis}}.$$

✗ In the MNAR setting, one should consider the mechanism.

How to deal with the MNAR data ?

- Modeling the MNAR mechanism via a **Logistic Model**:
  $\rightsquigarrow$ self-masked MNAR missing-data mechanism
  $\forall i \in [1, n]$, $\phi_j = (\phi_{1j}, \phi_{2j})$ denoting a parameter vector:

  $$p(M_{ij}|y_{ij};\phi) = [(1+e^{-\phi_{1j}(y_{ij}-\phi_{2j})})^{-1}]^{(1-M_{ij})}[1-(1+e^{-\phi_{1j}(y_{ij}-\phi_{2j})})^{-1}]^{M_{ij}}$$

- Maximizing the joint log-likelihood with the **EM algorithm** for instance in low-rank models [S., Boyer, Josse 2018].
  - ✓ Handling MNAR data in low-rank models.
  - ✗ Computationally costly.

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- Maximizing the joint log-likelihood with the **EM algorithm** for instance in low-rank models [S., Boyer, Josse 2018].
  - ✓ Handling MNAR data in low-rank models.
  - ✗ Computationally costly.
- Recent approach proposed by Pearl and Mohan (2018) based on graphical models and causality to handle the MNAR data in the linear models without modeling the mechanism and by only using observed data.

$\Rightarrow$ Proposal: Handling the MNAR data in probabilistic PCA model.

Probabilistic Principal Component Analysis (PPCA) model

$Y \in \mathbb{R}^{n \times p}$ noisy realisation of the factorization of the coefficients matrix $B \in \mathbb{R}^{r \times p}$ and r latent variables grouped in the matrix $W \in \mathbb{R}^{n \times r}$:

$$Y = \mathbf{1}\alpha + WB + \epsilon,$$

where

$$\begin{cases} W = (W_{1.}|\dots|W_{n.})^T, \text{ with } W_{i.} \sim \mathcal{N}(0_r, \mathrm{Id}_{r \times r}), \\ B \text{ with rank } r < \min\{n, p\}, \\ \alpha \in \mathbb{R}^p \text{ and } \mathbf{1} = (1 \dots 1)^T \in \mathbb{R}^n, \\ \epsilon = (\epsilon_{1.}|\dots|\epsilon_{n.})^T, \text{ with } \epsilon_{i.} \sim \mathcal{N}(0_p, \sigma^2 \mathrm{Id}_{p \times p}). \end{cases}$$

⇢ Access only to the missing-data matrix

$$Y \odot M + \mathrm{NA} \odot (1 - M)$$

Proposal

$$Y = \begin{array}{c} \begin{array}{cccccc} Y_{.1} & Y_{.2} & Y_{.3} & Y_{.4} & \dots & Y_{.p} \end{array} \\ \begin{pmatrix} 12 & 28 & 31 & 3 & \dots & \mathrm{NA} \\ \mathrm{NA} & 23 & 89 & 2 & \dots & 85 \\ 32 & 6 & 24 & \mathrm{NA} & \dots & \mathrm{NA} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ \mathrm{NA} & 3 & 7 & \mathrm{NA} & \dots & 11 \end{pmatrix} \end{array}$$

Under the PPCA model:

- Mean, variance and covariances estimation of the missing variables without explicitly modeling the MNAR mechanism and by only using observed data [**consistency results**],
- Estimation of the loading matrix B,
- Imputation of the missing values in the data matrix Y.

Graphical model

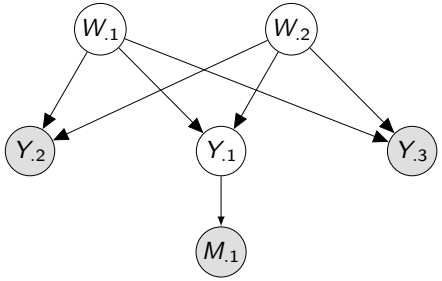

Figure: Graphical model for "fully-connected" PPCA model when $p = 3$, $r = 2$ and one variable $Y_{.1}$ is missing.

Mean, variance and covariances estimation

Algebraic approach

Toy example: $p = 3$, $r = 2$, one variable $Y_{.1}$ is missing.

$$\begin{pmatrix} Y_{.1} & Y_{.2} & Y_{.3} \end{pmatrix} = \mathbf{1} \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \end{pmatrix} + \begin{pmatrix} W_{.1} & W_{.2} \end{pmatrix} B + \epsilon$$

Lemma exploiting the linear links between variables

Assume that the reduced matrix $\begin{pmatrix} B_{.1} & B_{.3} \end{pmatrix}$ of B has an inverse matrix denoted as $B^{-13} \in \mathbb{R}^{2\times 2}$.

$$Y_{.2} = \mathcal{B}_{2\to 1,3[0]} + \mathcal{B}_{2\to 1,3[1]}(Y_{.1} - \epsilon_{.1}) + \mathcal{B}_{2\to 1,3[3]}(Y_{.3} - \epsilon_{.3}) + \epsilon_{.2},$$

where $(\mathcal{B}_{2\to 1,3[k]})_{k\in\{0,1,3\}}$ depends on the unknown matrix B:

$$\begin{aligned}
\mathcal{B}_{2\to 1,3[1]} &:= B_{11}^{-13} B_{12} + B_{12}^{-13} B_{22}, \\
\mathcal{B}_{2\to 1,3[3]} &:= B_{21}^{-13} B_{12} + B_{22}^{-13} B_{22}. \\
\mathcal{B}_{2\to 1,3[0]} &:= -\mathcal{B}_{2\to 1,3[1]} \mathbf{1}\alpha_1 - \mathcal{B}_{2\to 1,3[3]} \mathbf{1}\alpha_3 + \mathbf{1}\alpha_2,
\end{aligned}$$

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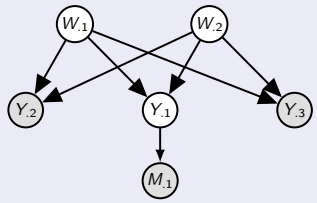
Graphical approach



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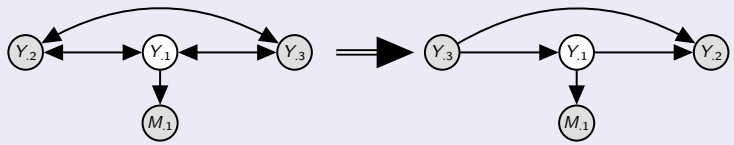

Coefficients standing for the effects of $Y_{.2}$ on $Y_{.1}$ and $Y_{.3}$ in the complete case when $M_{.1} = 1$:

$$(Y_{.2}|M_{.1} = 1) := \mathcal{B}^c_{2\to 1,3[0]} + \mathcal{B}^c_{2\to 1,3[1]}(Y_{.1} - \epsilon_{.1}) + \mathcal{B}^c_{2\to 1,3[2]}(Y_{.3} - \epsilon_{.3}) + \epsilon_{.2}.$$

Mean formula

Assume that:

A1. $\begin{pmatrix} B_{.1} & B_{.3} \end{pmatrix}$ is an invertible matrix,

A2. $Y_{.2} \perp\!\!\!\perp M_{.1} | Y_{.1}, Y_{.3}$ ($\rightsquigarrow$ ok for the self-masked MNAR mechanism).

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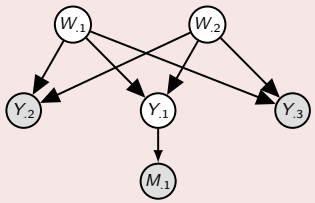

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Assuming also that $\mathcal{B}^c_{2\to 1,3[1]}$ is non-zero, one can derive that

$$\alpha_1 = \frac{\alpha_2 - \mathcal{B}^c_{2\to 1,3[0]} - \mathcal{B}^c_{2\to 1,3[3]}\alpha_3}{\mathcal{B}^c_{2\to 1,3[1]}}.$$

Consistency results

Definition of a mean estimator:

$$\hat{\alpha}_1 := \frac{\hat{\alpha}_2 - \hat{\mathcal{B}}^c_{2\to 1,3[0]} - \hat{\mathcal{B}}^c_{2\to 1,3[3]}\hat{\alpha}_3}{\hat{\mathcal{B}}^c_{2\to 1,3[1]}}.$$

Consistency for the missing variable mean

Assume that:

A3. α_2 and α_3 are recoverable, *i.e.* there exist consistent estimators for both quantities,

A4. the coefficients $\mathcal{B}^c_{2\to 1,3[0]}$, $\mathcal{B}^c_{2\to 1,3[1]}$ and $\mathcal{B}^c_{2\to 1,3[3]}$ are recoverable.

Then, the estimator $\hat{\alpha}_1$ is consistent.

Estimation in practice

$$\hat{\alpha}_1 := \frac{\hat{\alpha}_2 - \hat{\mathcal{B}}^c_{2\to 1,3[0]} - \hat{\mathcal{B}}^c_{2\to 1,3[3]}\hat{\alpha}_3}{\hat{\mathcal{B}}^c_{2\to 1,3[1]}}.$$

- $\hat{\alpha}_2$ and $\hat{\alpha}_3$ are computed as empirical quantities.
  - $\hat{\alpha}_2 = \bar{Y}_{.2}$
  - $\hat{\alpha}_3 = \bar{Y}_{.3}$

$$Y = \begin{array}{c} \begin{array}{ccc} Y_{.1} & Y_{.2} & Y_{.3} \end{array} \\ \begin{pmatrix} 12 & 28 & 31 \\ \mathrm{NA} & 23 & 89 \\ 32 & 6 & 24 \\ \vdots & \vdots & \vdots \\ \mathrm{NA} & 3 & 7 \end{pmatrix} \end{array}$$

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- $(\mathcal{B}^c_{2\to 1,3[k]})_{k\in\{0,1,3\}}$ estimated by the coefficients of the linear regression of $Y_{.2}$ on $Y_{.1}$ and $Y_{.3}$ using the fully-observed data only.

$$Y = \begin{array}{c} \begin{array}{ccc} Y_{.1} & Y_{.2} & Y_{.3} \end{array} \\ \begin{pmatrix} 12 & 28 & 31 \\ \cancel{\mathrm{NA}} & \cancel{23} & \cancel{89} \\ 32 & 6 & 24 \\ \vdots & \vdots & \vdots \\ \cancel{\mathrm{NA}} & \cancel{3} & \cancel{7} \end{pmatrix} \end{array}$$

- Same methodology for the variance and covariances.

Estimation of the loading matrix B

- Estimators obtained from the formulae:

$$\hat{\Sigma} = \begin{pmatrix} \widehat{\mathrm{Var}}(Y_{.1}) & \widehat{\mathrm{Cov}}(Y_{.1}, Y_{.2}) & \widehat{\mathrm{Cov}}(Y_{.1}, Y_{.3}) \\ \widehat{\mathrm{Cov}}(Y_{.2}, Y_{.1}) & \widehat{\mathrm{Var}}(Y_{.2}) & \widehat{\mathrm{Cov}}(Y_{.2}, Y_{.3}) \\ \widehat{\mathrm{Cov}}(Y_{.3}, Y_{.1}) & \widehat{\mathrm{Cov}}(Y_{.3}, Y_{.2}) & \widehat{\mathrm{Var}}(Y_{.3}) \end{pmatrix}$$

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.

- Assuming that σ^2 is known,

$$Y \sim \mathcal{N}\left(\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}, B^T B + \sigma^2 \mathrm{Id} \right) \Rightarrow \hat{\Sigma} - \sigma^2 \mathrm{Id}_{3\times 3} \text{ estimates } B^T B.$$

- Singular value decomposition:

$$\hat{\Sigma} - \sigma^2 \mathrm{Id}_{3\times 3} =: \hat{U} \hat{D} \hat{U}^T, \text{ with } \hat{U} = (\hat{u}_1 | \hat{u}_2 | \hat{u}_3).$$

- Assuming that $r = 2$,

$$\hat{B} = \hat{D}_{|2}^{1/2} \hat{U}_{|2}^T = \begin{pmatrix} \sqrt{\hat{d}_1} & 0 \\ 0 & \sqrt{\hat{d}_2} \end{pmatrix} \begin{pmatrix} \hat{u}_1^T \\ \hline \hat{u}_2^T \end{pmatrix}.$$

Imputation of the missing values in Y

- Impute the missing values Y_{i1} for $i \in \{1, \ldots, n\}$ such that $M_{i1} = 0$ using the conditional expectation of (Y_{i1}) given Y_{i2} and Y_{i3}.

$$\hat{A} = \hat{B}^T \hat{B} + \sigma^2 \mathrm{Id}_{3\times 3},$$

$$\hat{Y}_{i1} = \hat{\alpha}_1 + \begin{pmatrix} \hat{A}_{12} & \hat{A}_{13} \end{pmatrix} \begin{pmatrix} \hat{A}_{22} & \hat{A}_{23} \\ \hat{A}_{32} & \hat{A}_{33} \end{pmatrix}^{-1} \left(\begin{pmatrix} Y_{i2} \\ Y_{i3} \end{pmatrix} - \begin{pmatrix} \hat{\alpha}_2 \\ \hat{\alpha}_3 \end{pmatrix} \right).$$

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The methodology is extended to the general case, for any data with p covariates, r latent variables and d missing variables.

Numerical experiments

⋆ $N = 40$ simulations.

Measuring the performance:

- For the estimation of B: RV coefficient.
- For the imputation of Y: the prediction error,

$$\mathbb{E}\left[\left\|(\hat{Y} - Y) \odot (1 - M)\right\|_F^2\right] \Big/ \mathbb{E}\left[\|Y \odot (1 - M)\|_F^2\right]$$

.

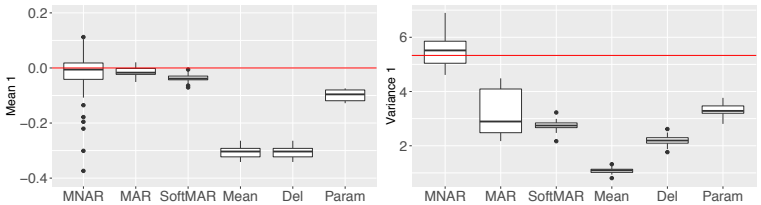

Figure: Mean and variance estimations of the missing variable $Y_{.1}$ when $r = 3$, $n = 1000$, $p = 10$, $\sigma = 0.1$ and 7 variables are missing leading to 35 % of MNAR values.

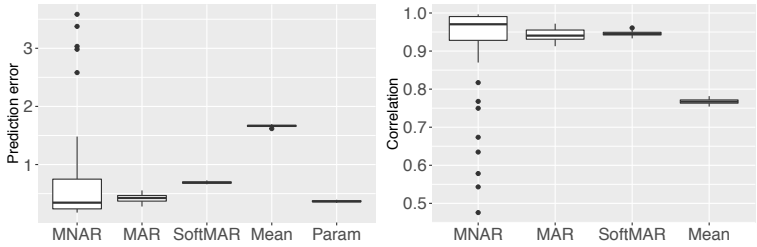

Figure: Prediction error for imputation and RV coefficient.

NB: Outliers due to the numerical non-stability of the matrix inversion.

Conclusion

Take-home messages

- MNAR is hard.
- Modeling the mechanism and using an EM algorithm is computationally costly.
- Our proposal consists in performing PPCA with MNAR data.
- **It does not require to model the mechanism.**

Perspectives

- Rank and noise level supposed to be known.
- Exogeneity approximation.
- Extended to other variants of PPCA.

[Sportisse et al., 2019]

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